Akan Selim

Distribution Steering • Robust System Design and Optimization • Stochastic Optimal Control • Nonlinear Programming • Astrodynamics

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Blackbird is still singing the tale of the mighty winter forest...

Aerospace PhD researcher at Georgia Tech and former **Senior GN&C Engineer at ROKETSAN** with experience across astrodynamics, trajectory optimization, launch-vehicle and spacecraft control, robust system design, stochastic controllers, and measure-valued methods. Current research focuses on **optimal transport, stochastic optimal control/stopping, distribution steering, nonlinear programming, probability measures over probability measures, and contraction metrics**. FieldAI gives me practical exposure to robotics AI, VLAs, large behavior models, Linux/Git/Docker workflows, and validation/verification. I aim to use control, optimization, and measure-valued probability tools for robotics autonomy, computational neuroscience, and understanding learning mechanisms in LLMs/LBMs.

CORE EXPERTISE

Optimal Transport | Stochastic Calculus | Diffusion Models

Astrodynamics | Trajectory Optimization

Stochastic / Robust / Optimal / Adaptive Control

MPC / MPPI | Path Integral Control

Learning-Based Optimal Control

Sparse Nonlinear Programming

Convex Optimization

Aerospace / Launch Vehicle System Design

Data-Driven Dynamic Learning | V&V

Linux | Git/GitHub | Docker | Python/PyTorch

EDUCATION

Georgia Institute of Technology 2024–Exp. 2029
PhD, Aerospace Engineering; Minor in Mathematics; **GPA: 4.0/4.0**.
Courses: Stochastic Calculus I–II, Optimal Transport Theory, Stochastic Optimal Control, Advanced Nonlinear Control, High-Dimensional Statistics, Hypersonic System Design.

Istanbul Technical University 2020–2022
MSc, Aeronautics and Astronautics Engineering; **GPA: 3.82/4.0**.
Thesis: Robust Re-entry Trajectory Optimization via Stochastic-Collocation Based Ensemble Pseudospectral Optimal Control.
Courses: Adaptive Control Theory, Deep Reinforcement Learning, Learning-Based Optimal Control, Optimization Theory.

Istanbul Technical University 2016–2020
BSc, Astronautical Engineering, Summa Cum Laude.

SKILLS

Programming: Python, PyTorch, JAX, NumPy/SciPy, MATLAB, Octave, Simulink, C.

Tools: Linux, Git/GitHub, Docker, pytest, CVX, IPOPT.

Math/ML: Optimal transport, diffusion/score models, stochastic calculus, probability measures over probability measures, sparse optimization, deep reinforcement learning, nonlinear programming, convex optimization.

Control: LQR, Lyapunov-based control, deep reinforcement learning control, MPPI, path-integral control, pseudospectral optimal control, structured H_∞, contraction metrics, robust/adaptive control.

Modeling: Data-based dynamical modeling, Koopman/Perron–Frobenius operators, polynomial chaos, Gaussian mixture models, unscented transformations, covariance intersection, sim-to-real V&V.

EXPERIENCE

FieldAI Irvine, CA, USA
Research Intern, Data-Driven Dynamics / VLA and Large Behavior Models Summer 2026

- Gaining practical exposure to **field foundation models, Vision-Language-Action models, large behavior models, diffusion-style policies, Linux/Git/Docker workflows, and validation/verification** in a robotics AI environment.
- Main technical angle remains **learning dynamics from data** for control/autonomy: rollout analysis, capability/failure-mode evaluation, robustness validation, and sparse/optimization-based interpretability of policies.

Georgia Institute of Technology Atlanta, GA, USA
Graduate Research Assistant, Stochastic Optimal Control / Robotics Autonomy Aug. 2024–Present

- Developing theory and computation for **stochastic optimal control, distribution/covariance steering, optimal stopping, Fokker–Planck transformations, nonlinear optimal transport, and free-final-time distribution steering**.
- Studying **probability measures over probability measures**, measure-valued dynamics, stochastic calculus, and learning-based approximations for robust planning/control under uncertainty.
- Building MPPI/path-integral algorithms combining costate information, sparsity, optimal transport, and nonlinear programming for robot autonomy and trajectory optimization.

Georgia Institute of Technology Atlanta, GA, USA
Teaching Assistant, AE 4610: Dynamics and Control Laboratory Spring 2026

- Taught two laboratory sections covering **reaction-wheel spacecraft attitude control, inverted pendulum stabilization, system identification, and real-time controller implementation**.
- Guided students on **PID, LQR, Lyapunov-based control, and deep reinforcement learning control**, connecting controller synthesis, tuning, robustness checks, and experimental validation.

ROKETSAN Inc. Istanbul, Turkiye
Senior Guidance, Navigation & Control Engineer Jan. 2021–Aug. 2024

- Developed launch-vehicle **multidisciplinary optimization** and trajectory/configuration design software in MATLAB/Octave/Python using convex optimization, sparse nonlinear programming, pseudospectral optimal control with sparse Hessian approximations, UQ margins, PCE/unscented filtering, Monte Carlo validation, and hybrid optimizers.
- Designed and verified real-time guidance/control algorithms for orbital insertion, rendezvous/docking, re-entry, and spacecraft mission design: IGM/UPEG, Lambert solvers, robust autopilots, structured H_∞, and contraction-metric controllers.
- Built automated log-analysis/replay tools, vectorized Monte Carlo validation, unit-tested simulation infrastructure, optimal control allocation, Sinkhorn/optimal-transport modules, and Koopman/Perron–Frobenius data-driven dynamics models for control design and verification.

Akan Selim – Research Portfolio and Publications

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PROJECTS / RESEARCH PORTFOLIO

- **Path Integral Control and MPPI:** Sparse MPPI for dual control inside MPPI using sparse kernel functions and adaptive feature-weight updates; optimal-transport-based MPPI; costate-informed rollouts exploiting optimality structure through costate dynamics.
- **Constrained Distributional Optimization:** Double-sided path-integral methods for gradient-free nonlinear programming under distributional constraints; cost-informed MPPI for optimal-transport optimal control applied to spacecraft formation reconfiguration in elliptical orbit.
- **DNN Warm-Start Pseudospectral Control:** Costate-dynamics warm starts using covector mapping theory and ensemble optimal control; checkpoint logic for switching from robust to deterministic optimal-control logic.
- **Robust Policy Learning:** Neural-network policies for integrated guidance/control under uncertainty, including translational and rotational policy modules, reachability checkpoints, and robust recursive replanning.
- **Measure-Valued Stochastic Control:** Distribution steering, covariance steering, stochastic stopping, Fokker–Planck transformations, and nonlinear programming over probability laws for uncertainty-aware autonomy.

SELECTED TECHNICAL RESULTS

- Built optimization-driven launch-vehicle configuration/trajectory design software with realistic uncertainty margins and Monte Carlo validation.
- Developed hybrid gradient-based and gradient-free optimizers in MATLAB/Octave/Python for mission/system design, nonlinear programming, and robust trajectory optimization.
- Used polynomial chaos, unscented filtering, ensemble pseudospectral optimal control, and replay/log analysis to stress-test closed-loop guidance algorithms.
- Built control-oriented model-learning pipelines using Koopman/Perron–Frobenius ideas, data-driven dynamics, and validation/verification loops.
- Integrated sparse nonlinear programming, sparse Hessian approximations, and multi-phase optimal-control solvers for reusable trajectory-optimization infrastructure.

PROJECTS

- **ASELSAN collision avoidance:** Collision early-warning and avoidance systems for N-to-N analysis with robust and optimal maneuver design.
- **ASELSAN re-entry hypersonics CFD validation:** Aerodynamic/heating validation context for trajectory optimization and control studies.
- **Mission/system design:** Apophis microsatellite concept and expandable lunar-base concept; system-level analysis and multidisciplinary trade studies.

RESEARCH DIRECTION

- Robotics autonomy, interpretable large behavior models, data-driven dynamics, robust policy learning, VLA evaluation, stochastic control, measure-valued dynamics, nonlinear programming, uncertainty-aware planning, and safety V&V.

SELECTED PUBLICATIONS / MANUSCRIPTS

- **Continuous-Time Covariance Steering with Common Free-Final Time, 2026**: Theory for turnpikes used in distribution steering problems, including phenomena that appear in optimal policies across small- and large-time-horizon regimes.
- **Nonlinear Stochastic Optimal Control and Optimal Stopping using Fokker–Planck Transformation, 2026**: Theory for stochastic control, distribution steering, optimal stopping, common optimal free-final-time transversality conditions, and diffusion/score-based control.
- **Stochastic Trajectory and Controller Optimization via Optimal Recovery Diffusion Control, 2024**: Diffusion-control viewpoint for stochastic optimal recovery policies for a Mars re-entry vehicle using score matching via sparse Gaussian kernel functions.
- **Stochastic Trajectory and Robust Controller Optimization via Contractive Optimal Control, 2024**: Integrated guidance and control architecture that designs both the reference trajectory and robust controller using control contraction metrics inside nonlinear programming.
- **Stochastic Optimal Control under Non-Gaussian Uncertainties via Entropy Minimization and Dynamical Indicators, 2024**: Entropy-based uncertainty treatment, nonlinear stochastic dynamics, and robustness indicators for safety-critical autonomy.
- **Real-Time Optimal Control of a 6-DOF State-Constrained Close-Proximity Rendezvous Mission, Acta Astronautica, 2023**: Ensemble optimal control, warm-start Bellman pseudospectral control, DNN costate-policy learning, reachability checkpoints, and robust recursive replanning under estimation errors.
- **Robust Trajectory Optimization of Re-entry Flight with Prescribed Endpoint Region via Sparse Grid Ensemble Pseudospectral Optimal Control, Advances in Space Research, 2023**: Sparse-grid uncertainty propagation, ensemble pseudospectral optimization, endpoint-region constraints, and robustness against dispersions.
- **EPOCS / Modular Multi-Phase Pseudospectral Optimal Control Solver, 2023**: Software-oriented optimal-control work covering phase linkage, constraints, warm starts, solver validation, and reusable trajectory-optimization infrastructure.
- **Recovery Ensemble Control; Desensitized Ensemble Guidance; Multi-phase Robust Trajectory Optimization of a Hybrid Guidance Architecture for Launch Vehicles, 2023**: Robust guidance, integrated trajectory/control design, uncertainty-aware optimization, and verification of safety-critical autonomy.
- **Integrated Guidance and Fuel-Optimal Control of a Space Servicing Vehicle via Warm-Start Bellman Pseudospectral Method, 2022**: Learning robust policies through neural networks, costate/covector warm starts, modular optimal-control software, and nonlinear programming.